Spool Film Processor – schematics and assembly

ESP32-C3 + 2.4" TFT (msalamon EVB-00055) · TMC2209 · JK42HS48-1204 · DS18B20 · XL6009 · USB-C charger module · Li-ion 5000 mAh · LM386 + speaker · **firmware v2.2** · 2026-10-16

1. BOM 2. Main schematic 3. Speaker 4. Identifying parts 5. Wiring tables 6. Assembly order 7. Seals 8. Flashing 9. Usage

> **Rule:** the board's header exposes only **GPIO21 (TXO)** and **GPIO20 (RXO)**. The motor is driven over one wire (UART) and the probe over the other (1-Wire). The **BOOT (GPIO9)** button works as START and also feeds the speaker. Everything else is power.

1. BOM – bill of materials

Valid for firmware **v2.2**. "Donor board" = an old board (here AK-750S) the small SMD parts were desoldered from. All resistors and capacitors can also be bought new (through-hole); the values are the same.

A. Modules

Ref.	Part	Type / parameters	Qty	Notes
MCU	ESP32-C3 board with display	msalamon **EVB-00055**, 2.4" ST7789 240×320 TFT, CST816 touch	1	only GPIO20, GPIO21 free (+ BOOT = GPIO9)
DRV	stepper driver	TMC2209 **Tenstar V985**	1	+ heatsink on the chip
M1	stepper motor	**JK42HS48-1204** (NEMA17, 1.2 A, 5 mm shaft with a full-length D-flat)	1	runs at 600–700 mA
TS	temperature probe	**DS18B20** waterproof, 3 m cable	1	red VCC, yellow DATA, black GND
BST	step-up converter	**XL6009**, set to 12.0 V	1	set the voltage before connecting the driver
CHG	charger + 5 V module	USB-C, **5 V 4 A** output	1	charges the battery and supplies 5 V
BAT	battery	Li-ion 3.7 V **5000 mAh**	1	about 3 h of continuous agitation at 700 mA
SW1	power switch	toggle / rocker, **≥ 3 A**	1	between the charger module and the rest
SP1	speaker	**14 Ω** PC case speaker	1	red + black wire

B. Electronic parts

Ref.	Value	Function	Qty	Marking
R1	4.7 kΩ	DS18B20 line pull-up to 3.3 V	1	SMD **4701** · through-hole: yellow-violet-red
R2	1 kΩ	GPIO21 → driver PDN (UART)	1	SMD **1001** · brown-black-red
R5	10 kΩ	LM386 input divider (series)	1	SMD **1002** · brown-black-orange
R6	470 Ω	LM386 input divider (to GND)	1	SMD **4700** · yellow-violet-brown
R7	**100 Ω** ≥ 0.25 W	LM386 output → speaker (**replaces the former C4 100 µF**)	1	SMD **1000** / **101** · brown-black-brown
C1	1000 µF 25 V	12 V motor supply buffer at VM	1	large through-hole

C2	2.2 µF 50 V	GPIO9 → LM386 coupling (blocks DC)	1	SMD can "2.2 50V"
C3	47 µF 25 V	LM386 supply filter (pin 6)	1	SMD can "47 25V"
C5	10 µF 50 V	LM386 pin 7 – less noise	1	SMD can "10 50V"
C6	10 µF 50 V	LM386 gain 200× (pins 1–8)	1	SMD can "10 50V"
U1	LM386 + DIP-8 socket	speaker amplifier	1	"LM386" on the chip

C. Assembly materials

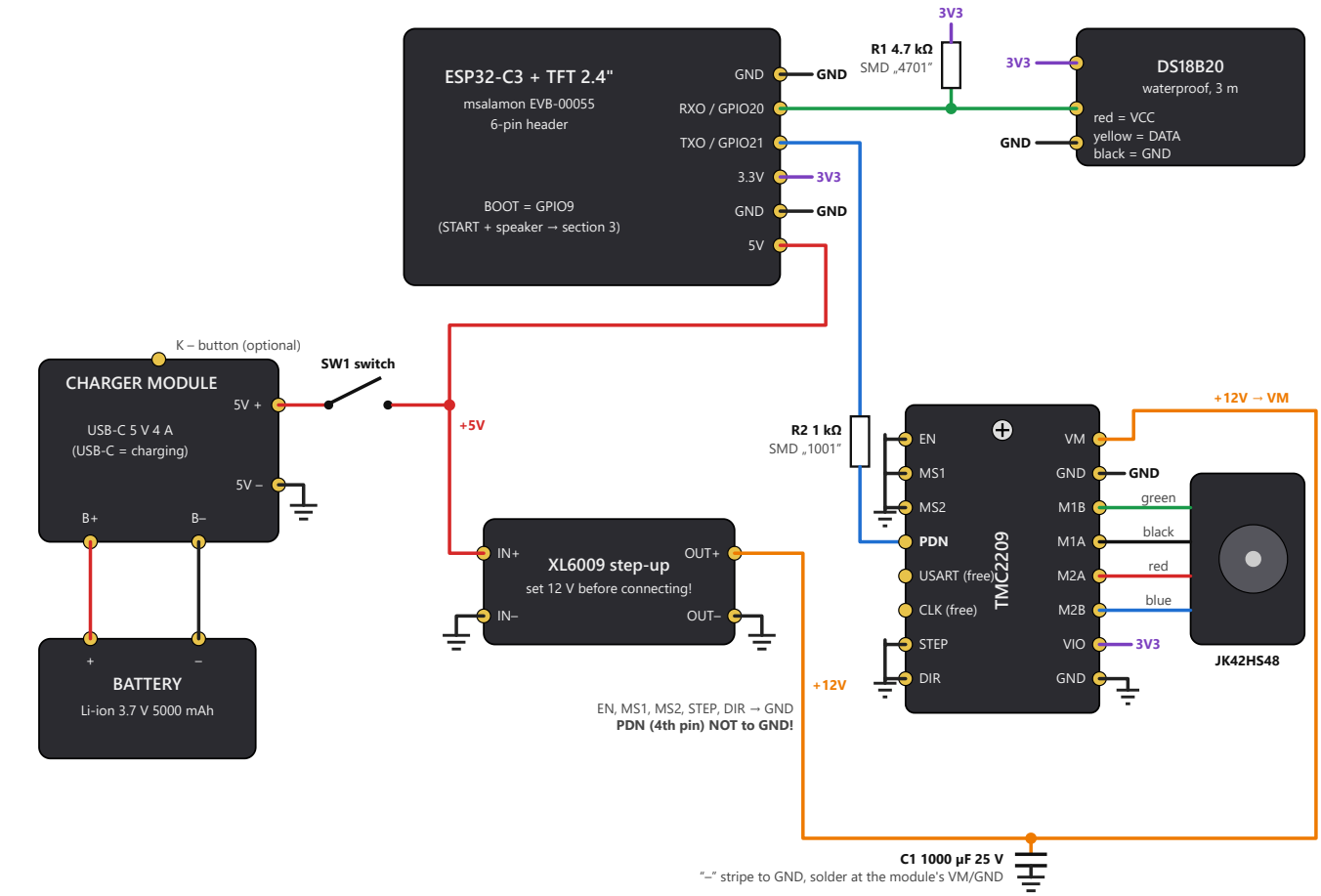
Item	Parameters	Qty	Notes
perfboard	approx. 5×7 cm	1	for the LM386 and small parts
power wires	silicone 18–20 AWG	~1 m	5 V, 12 V, motor GND
signal wires	22–26 AWG	~2 m	UART, 1-Wire, audio, 3.3 V
pin headers / JST connectors	2.54 mm pitch	as needed	so the modules can be unplugged
heat-shrink tubing	with adhesive	set	also for the motor cable exit
heatsink	for the TMC2209	1	thermally glued

D. Mechanics and seals (section 7)

Item	Parameters	Qty	Where
O-ring	**131×3 NBR70** (or 131×3.5)	1 + spare	groove that takes the Paterson tank rim
Shaft seal	**8×16×5 TC**, preferably **FKM** (NBR also works)	2	shaft passing through the front (one spare)
Sleeve	stainless tube **8×1.5 mm** (5 mm ID), 12–15 mm long	1	on the motor shaft – a round surface for the seal
epoxy	2-part	1	glues the sleeve on (fills the D-flat)
silicone grease	—	1	seal lip, O-rings
bearing (optional)	688-2RS (8×16×5)	1	in the same Ø16 bore behind the seal
filament	**PETG** (front and wet parts), **PLA-CF** (back)	—	PETG resists the chemistry and 38–45 °C
alternative	neodymium magnets 10×5 mm N42–N52	8	magnetic coupling instead of a shaft seal

> Measure every resistor taken from a donor board with a multimeter before use. 3D print files will be published on MakerWorld.

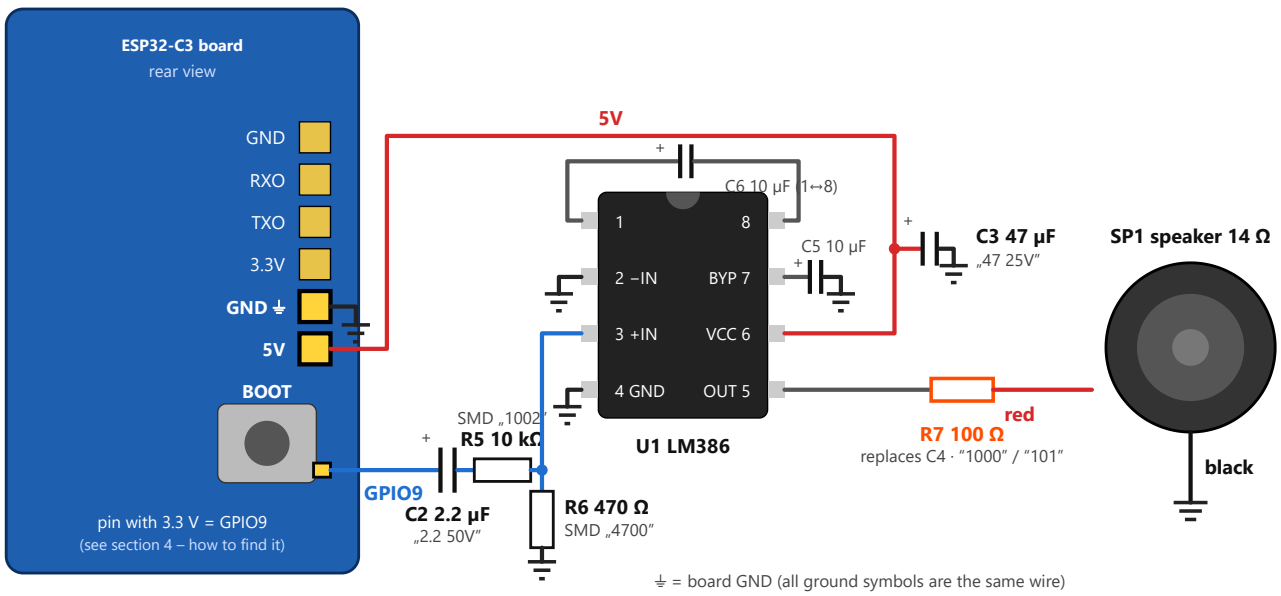
2. Main schematic



+5 V · +12 V (motor) · 3.3 V from the ESP board · GND – all grounds together · UART to TMC2209 · 1-Wire DS18B20

The ⏚ symbol / GND label = common ground, 3V3 label = the board's "3.3V" pin. These connections are shown as labels instead of drawn wires so that lines do not cross.

3. Speaker through the LM386 amplifier (on the BOOT button)



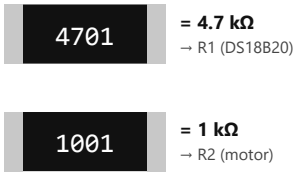
Step by step

1. Solder the **8-pin socket** to the perfboard. Insert the LM386 last.
2. **Pin 6 (VCC)** → the board's **5V** pin. Capacitor **C3 47 µF**: plus to pin 6, minus to GND, as close to the chip as possible.
3. **Pins 2 and 4** → **GND**.
4. **GPIO9** (the BOOT pin with 3.3 V) → **C2 2.2 µF** (plus towards GPIO9) → **R5 10 kΩ** ("1002") → **pin 3**.
5. **R6 470 Ω** ("4700") between **pin 3 and GND**. R5 and R6 form a divider that reduces the 3.3 V signal to about 0.15 V, which the LM386 accepts without clipping.
6. **C6 10 µF** between **pin 1 (+) and pin 8 (–)**. Raises the gain from 20× to 200×.
7. **Pin 5 (OUT)** → **R7 100 Ω** → **red** speaker wire. **Black** → GND. R7 has no polarity. *(R7 replaces the former C4 100 µF capacitor.)*
8. **C5 10 µF**: plus to pin 7, minus to GND. Recommended at high gain, it lowers the noise.
9. Insert the LM386 into the socket with the **notch on the pin-1 side**.

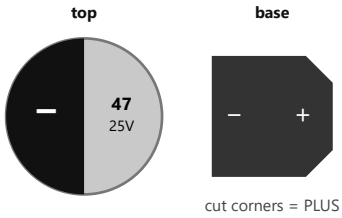
How it works

- The firmware briefly outputs an audio signal of about 1.5–4 kHz on GPIO9. The LM386 amplifies it and drives the speaker.
- **C2** passes only the audio to the amplifier and blocks DC, so the ESP board still boots normally (GPIO9 must be high at power-on).
- **R7 100 Ω** limits the speaker current. About **20 mA** of DC flows through the coil (about 7 mW) – safe for the 14 Ω speaker and the amplifier. In this build a 100 µF capacitor in this position sounded too quiet, so R7 is used.
- The BOOT button still works as **START / PAUSE / NEXT**; pressing it gives a quiet click.
- **RESET** (labelled 复位) is just a hardware CPU restart.

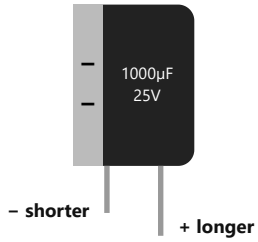
4. Identifying parts



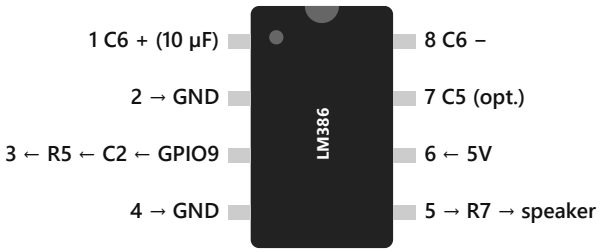
SMD resistor: the first 3 digits are the value, the fourth is the number of zeros. 470 + 1 zero = 4700 Ω = 4.7 kΩ; 100 + 1 zero = 1000 Ω = 1 kΩ. Polarity does not matter.



SMD can: black field on top = MINUS (−). On the base, the cut corners are on the plus side.

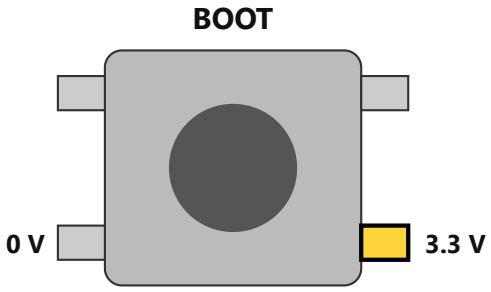


Through-hole capacitor (C1 1000 µF): **light stripe with "−" = MINUS**. After desoldering both legs are equal, so go by the stripe.



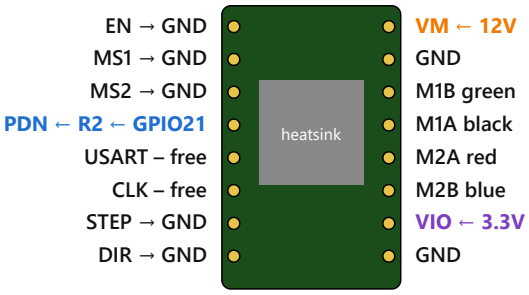
top view, notch at the top

LM386: the notch or dot marks pin 1. Numbering runs down the left side (1–4) and back up the right side (5–8).



example – your pin may differ, always check with a multimeter

GPIO9 pin: with the board on USB, measure DC voltage between the BOOT pins and GND. The pin with **~3.3 V** is GPIO9, the one with **0 V** is ground. Unplug USB before soldering.



TMC2209 V985 (verified: answers at address 0). Pinout per the manufacturer (Tenstar). R2 goes to **PDN** (4th pin); **USART** (5th pin) stays free. Without 3.3 V on **VIO** the driver does not answer and pulls the PDN line low.

5. Wiring tables

ESP32-C3 board (6-pin header + BOOT)

Board pin	Connect to	Notes
5V	+5V after switch SW1 · LM386 pin 6	powers the ESP, display and amplifier
GND (both)	common ground	
3.3V	DS18B20 red · R1 4.7 kΩ · TMC2209 **VIO**	3.3 V logic
TXO / GPIO21	R2 1 kΩ (SMD 1001) → TMC2209 **PDN**	the only motor control wire
RXO / GPIO20	DS18B20 yellow + R1 4.7 kΩ (SMD 4701) to 3.3V	1-Wire
BOOT (GPIO9)	C2 2.2 µF (+ towards GPIO9) → R5 10 kΩ → LM386 pin 3 (and R6 470 Ω from pin 3 to GND)	solder to the button pin with 3.3 V

TMC2209

TMC2209 pin	Connect to	Notes
VM	+12 V from XL6009 OUT+ and C1 1000 µF (+)	C1 as close to the driver as possible
GND (next to VM)	GND, C1 (–)	thick wire
VIO	3.3V from the ESP board	UART does not work without it
GND (next to VIO)	GND	
PDN (NOT "USART"/"RX" – on the V985 that pad is not bonded!)	R2 1 kΩ → GPIO21 (TXO)	control wire
EN, MS1, MS2, STEP, DIR	GND	MS1/MS2 = address 0
USART	**not connected**	not bonded to the chip on the V985
CLK	not connected	
M1A, M1B	motor: **black → M1A, green → M1B** (coil 1, ~3.3 Ω)	
M2A, M2B	motor: **red → M2A, blue → M2B** (coil 2, ~3.3 Ω)	

USB-C charger module

Pad	Connect to
B+ / B– (next to "K")	battery + / –
5V + / 5V –	5V+ → SW1 → +5V, 5V– → GND

XL6009 and motor

Item	Connection
XL6009 IN+ / IN–	+5V (after SW1) / GND
XL6009 OUT+ / OUT–	TMC2209 VM / GND

K	optional: button to GND

Before connecting the battery, plug in USB-C and press K. The 5V pads should read ~5.1 V and the battery pads ~4.2 V.

Coil A	black + green (~3.3 Ω)
Coil B	red + blue (~3.3 Ω)

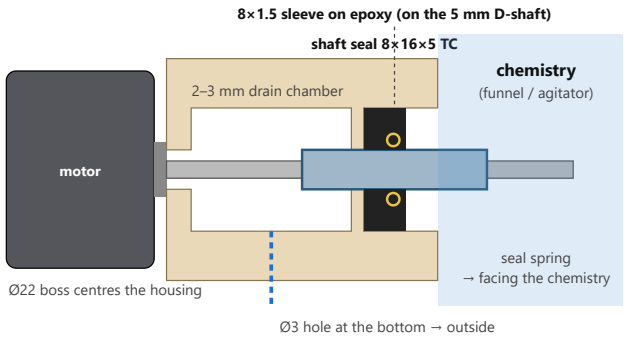
Wrong direction: swap the 2 wires of one coil or set `MOTOR_INVERT true`.

6. Assembly order

1. **Desolder the parts from the donor board** and measure the resistors.
2. **ESP board on the PC's USB**: flash the firmware (section 8). On first boot tap the logo and calibrate the screen.
3. **Speaker + LM386** (section 3): a melody plays at power-on and touching the screen gives a click.
4. **DS18B20 + R1** → Menu → Thermometer shows the temperature.
5. **Charger module + battery**: check polarity and 5 V at the output.
6. **XL6009 on its own**: feed 5 V and, without load, set **12.0 V** with the potentiometer.
7. **TMC2209 without the motor**: VIO (3.3 V), GND, VM (12 V), C1, **PDN** via R2 to GPIO21, and EN, MS1, MS2, STEP and DIR to GND. USART and CLK stay free. Check that nothing gets hot.
8. **Motor**: connect and disconnect only with power off. Then Settings → Motor test.
9. Finally switch SW1, seals (section 7) and mounting in the enclosure.

> **Power:** the module delivers up to 4 A at 5 V. The motor at 700 mA (stealthChop) draws about 0.6–1 A from 5 V, the ESP with the display about 0.1 A and the LM386 about 0.1 A while beeping, so there is plenty of headroom. Stick a heatsink on the TMC2209.

7. Seals and mechanics



Shaft passage: the shaft has a D-flat along its whole length, so the seal runs on an epoxied **8×1.5 stainless sleeve** (the epoxy fills the flat). Print the Ø16 bore at **Ø15.8–15.9 mm**, 5 mm deep, 0.5 mm entry chamfer. Behind the seal a chamber with a drain hole: if anything leaks past, it does not reach the motor bearing.

Shaft seal – rules

- **Lip spring towards the chemistry.** Mounted the other way round it leaks.
- Grease the lip and the sleeve with **silicone grease**. Rinse the area with water after each process: dried blix wears the lip.
- Centre the housing on the motor's **Ø22 mm boss** (ring Ø22.1–22.2 mm). An extra bearing is then not required; optionally a **688-2RS** in the same Ø16 bore.
- The motor itself is **not sealed** (ZZ bearing, housing joints). Seal the joints thinly with silicone and the cable exit with adhesive heat-shrink.
- Seal friction is about 0.02–0.05 Nm. After assembly check **Settings → Motor test**.

Tank O-ring

Dimension	Value
groove (measured in Fusion 360)	Ø130 – Ø137.7 mm, approx. 3.85 mm wide
O-ring	**131×3 NBR70** (or 131×3.5)
groove depth	**2.2–2.4 mm**, flat bottom
continuity	**full 360°**: move the drain cut-out inside the circle or run the groove over it

Alternative without a shaft seal: magnetic coupling – 2 discs Ø30 mm, 4 magnets 10×5 mm each (N-S-N-S) on Ø20 mm, 2 mm wall, magnet gap ≤ 4 mm. The magnets on the wet side must be fully embedded in PETG.

8. Flashing the firmware (v2.2)

`firmware` folder:

File	Purpose	Settings, calibration, programs
`SpoolFilmProcessor_v2.2_update.bin`	update (USB or WiFi), address 0x10000	**kept**
`SpoolFilmProcessor_v2.2_full.bin`	new or erased board, address 0x0	**erased**
`flash_update.bat`, `flash_full.bat`	one-click flashing (Windows)	—
`SpoolFilmProcessor_v2.2_ArduinoIDE.zip`	source code for Arduino IDE + instructions	**kept**
`esptool.exe`	Espressif flashing tool used by the .bat files – download `esptool-windows-amd64.zip` from github.com/espressif/esptool/releases	—

A. Over WiFi (no cable)

1. On the device: **Settings** → **WiFi update**.
2. Connect a phone or laptop to the **FilmProcessor** network, password **film1234**.
3. Open **http://192.168.4.1**.
4. Choose `SpoolFilmProcessor_v2.2_update.bin` and send it.
5. The device restarts by itself.

B. Over USB – update

1. Connect the board with a USB-C cable.
2. Run `flash_update.bat`.
3. Enter the port (Enter = COM3; check the number in Device Manager → Ports).
4. Wait for "DONE".

If the port does not connect: hold **BOOT**, press **RESET**, release BOOT and run it again.

C. New board – full flash

1. Run `flash_full.bat` and confirm with **Y**.
2. *Or* in Chrome or Edge open **espressif.github.io/esptool-js**, Connect, address **0x0**, file `SpoolFilmProcessor_v2.2_full.bin`, Program.
3. After boot **tap the logo** and calibrate the screen (3 crosses → DONE).

The flashed version is shown at the bottom right of the logo screen ("v2.2") and in **Settings** → **Info** (with the build date). Source code: the `SpoolFilmProcessor` folder (Arduino IDE, esp32 package 3.x – see INSTRUCTIONS.md).

9. Usage – quick guide

Starting a process

1. After power-on the device opens the **last used program** (back → list → process types).
2. **START** screen: processor upright / lying, **does the probe touch the chemistry**, chemistry volume, pre-wash temperature → **NEXT**.
3. On the first "POUR IN" screen the top shows **roll x/y** of chemistry usage and the bottom the estimated time of the whole process.
4. **START** on the screen or the **BOOT** button (START / PAUSE / NEXT).

Probe not in chemistry

- The room temperature is stored once, on NEXT.
- During pre-wash and rinses the probe works as a **tap-water thermometer**: drop of ~1 °C within a few seconds = water poured out (tank cooling counted from then), rise = refilled, max, "steady". Colours: blue < 38 °C, green 38–45 °C, red > 45 °C.
- Beeps: **too hot** – fast and high, **too cold** – slow and low.
- The chemistry temperature is computed by the thermal model of the tank (Paterson Multi Reel 3) and the time is extended accordingly.

Useful

- **Touch calibration:** touch the screen while the logo is shown. It is saved only after **DONE** or BOOT; without confirmation within 20 s it starts over.
- **Rotate screen 180°:** Settings → Rotate screen.
- **Chemistry usage:** on the DONE screen set the number of rolls → SAVE & FINISH. Developer and blix times grow per the manufacturer table.
- **Rinse by water changes:** step editor, "Water changes" field.
- **Upright tank – streaks at the bottom edge:** Settings → ROLL agitation: intense start 30 s, burst every 30 s (4 reversals). Soft start protects the agitator. An empty reel at the bottom keeps the film away from the exhausted layer at the tank floor.
- A pre-wash with **40–43 °C** water is safe for the emulsion and warms the tank. Avoid water above 45 °C and big temperature jumps between baths.